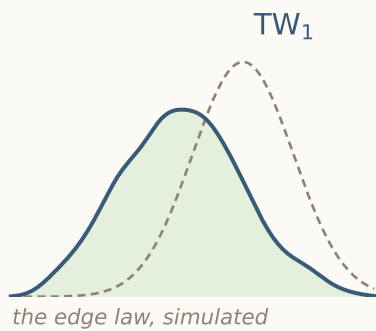
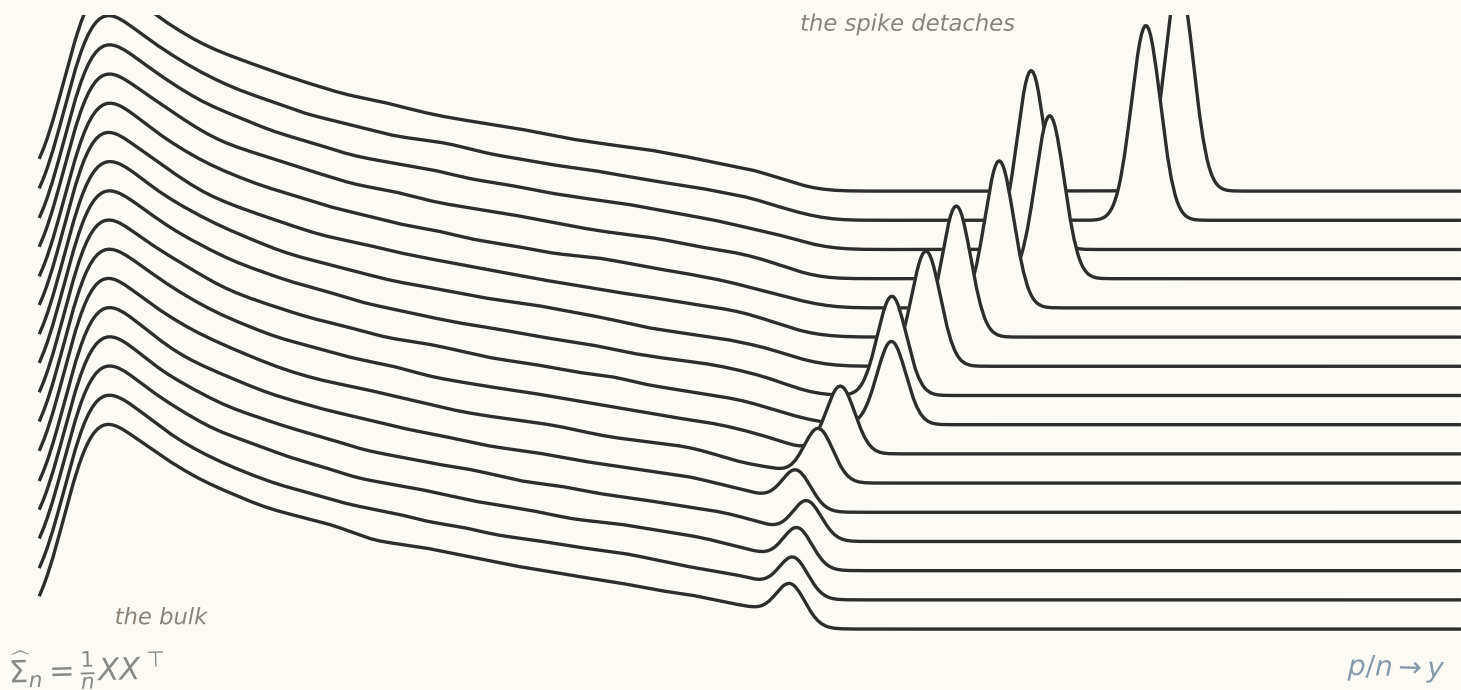
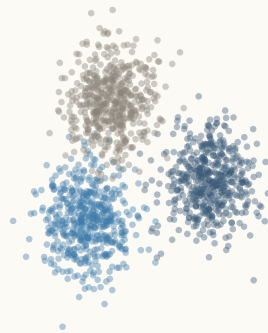




$$\lambda_{\pm} = (1 \pm \sqrt{y})^2$$



the spectral embedding

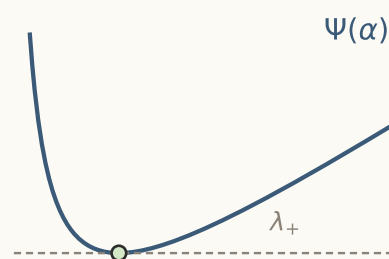
$$\alpha_c = 1 + \sqrt{y}$$

$$\rho(\alpha, y) = \frac{\alpha \Psi'(\alpha)}{\Psi(\alpha)}$$

$$\text{accuracy} \approx \Phi(\sqrt{\zeta/(1-\zeta)})$$

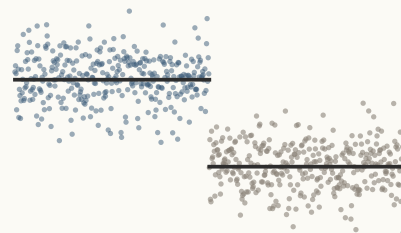
$$X = MJ^\top + W$$

$$\langle v_k, \widehat{v}_k \rangle^2 \rightarrow \zeta_k$$



the spike-location map

$$G = \frac{1}{p} X^\top X$$



the eigenvector knows the classes